

Week 5:Report Summary.

- **Problem Statement:** Predict Titanic survival using advanced ML.
- **Dataset:** Cleaned Titanic dataset (preprocessed in Week 3).
- **Models Used:** Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Tuned Random Forest.
- **Evaluation Metrics:** Accuracy, Confusion Matrix.
- **Performance Comparison:** Tuned Random Forest achieved the highest accuracy.
- **Recommendation:** Use ensemble methods with tuning for better predictive performance.